Production probability, transport, and grammatical possibility

Reanalysing the English dative alternation with open data

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Abstract

This paper develops a no-new-data reassessment of Bresnan et al. (2007). The current analyses reproduce the production-choice target from `languageR::dative`, add explicit null and partial-pooling checks, and test whether the resulting conditioning profile transports to the Spoken British National Corpus 2014 (BNC2014) dative dataset. It then adds a compact DAIS acceptability bridge for the same high-frequency verb core. The working result is that modern modelling preserves the classic production conclusion, while BNC2014 provides positive but partial cross-corpus transport evidence and DAIS shows moderate production/preference alignment with informative divergences. It reads that result through projectibility (what observing the conditioning profile licenses us to predict elsewhere): within the shared high-frequency verb set, the profile forecasts realization in a later spoken BNC2014 corpus-domain target and has a corresponding judgement profile. Production frequency stays silent, by construction, about the grammatical-but-unattested region, so the paper keeps production probability, acceptability preference, and grammatical possibility apart.

Keywords: dative alternation; production probability; transport; projectibility; grammaticality; corpus linguistics; acceptability

1 The problem

The English dative alternation is a small problem with a large methodological payoff. Speakers can say "give Kim the book" or "give the book to Kim", but the choice is uneven. It varies with verb, length, animacy, definiteness, pronominality, discourse status, corpus, and register.

That unevenness is easy to misread. A rare pattern can be lexically restricted, strongly dispreferred, or tied to a discourse context without falling outside the grammar. A production model therefore has two jobs. It has to predict attested choices, and it has to say what kind of grammatical claim those choices support.

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Bresnan et al. (2007) made the case that corpus probabilities can reveal grammatical possibilities that informal judgements miss. This paper asks how that case looks after open-data replication, modern validation, and cross-corpus transport.

The question is one of PROJECTIBILITY: what observing a conditioning profile in one setting licenses us to predict about another (Goodman, 1955). For the dative alternation, the relevant settings are a packaged reanalysis dataset, a later spoken British corpus, and possible acceptability evidence. Keeping those settings apart is the paper's main discipline.

2 The original target

Bresnan et al. (2007) is the anchor because it treats the alternation as a grammatical problem rather than as a simple frequency count. The claim is not that whatever occurs often is grammatical. The claim is that observed choices can expose systematic constraints on what speakers treat as available.

That move matters because informal judgements often compress several facts into one verdict. A construction may sound odd because it's impossible, because it is rare, because the wrong verb is present, because a discourse condition is missing, or because the constructed example strips away the context that would license it. Corpus data can help separate these cases.

The empirical target is production choice. In Bresnan et al.'s setup, the response variable is whether the recipient appears as a noun phrase (NP) in the double-object pattern or as a prepositional phrase (PP) in the prepositional dative. The predictors describe properties of the verb, recipient, theme, discourse status, and corpus source.

This paper preserves that target. It treats production probability as evidence to be interpreted, rather than as a direct measure of grammaticality. The empirical question is whether the production profile survives cleaner modelling and whether it travels beyond the original corpus setting.

3 Data and direct replication

The analysis starts with `languageR::dative`, distributed with the Comprehensive R Archive Network (CRAN) `languageR` package (Baayen & Shafaei-Bajestan, 2025). The source tarball supplies the packaged `dative.rda` object; the analysis scripts load it and write derived summaries.

The package documentation identifies the sources as Switchboard and the Treebank Wall Street Journal collection. Those sources place the original data in American English: Switchboard is telephone conversation among speakers from the United States, developed in 1990–1991 (Godfrey & Holliman, 1993), and the Treebank Wall Street Journal component is late-1980s American news and financial writing, with one million words of 1989 material and a larger 1987–1989 Wall Street Journal source collection (Charniak et al., 2000; Marcus et al., 1999).

In the packaged dataset, there are 3,263 rows and 15 columns. The response records recipient realization, with 2,414 NP realizations and 849 PP realizations. The dataset contains 75 verb levels and two modalities: 2,360 spoken tokens and 903 written tokens. Speaker identifiers are missing exactly for the written rows.

For the first model sequence, the response stays binary: NP is coded as 1 and PP as 0. The initial split is verb-stratified, with a fixed seed (`20260623`), so every verb observed in the test set is also represented in training. That split gives 2,637 training rows and 626 test rows.

The denominator is therefore narrow. These rows are already tokens in which a dative recipient was realized as NP or PP. They aren't all contexts in which a transfer event could have been expressed, omitted, paraphrased, or packaged in some other construction. The models estimate conditional choice within an attested alternation frame: given that a token enters the dative-alternation sample, which realization appears?

Initial analyses are deliberately plain. A marginal model sets the baseline. Structured models then add modality, length, verb, and the non-verb predictors. Two falsification checks accompany them: a scrambled-label check and fake-data checks. These tests ask whether the fitted structure beats chance, not whether it proves a grammatical theory.

A second empirical source is the Spoken British National Corpus 2014 (BNC2014) dative dataset (G. Jenset & McGillivray, 2018; G. B. Jenset & McGillivray, 2019). The verified Figshare comma-separated values (CSV) file has 1,839 parsed data rows and 44 columns under a Creative Commons Attribution 4.0 International (CC BY 4.0) licence. The apparent line-count discrepancy is mechanical: the file has 1,839 newline characters but no terminal newline, giving 1,840 logical text lines including the header.

The BNC2014 dative file comes from the Early-Access Subset of the Spoken BNC2014, whose spontaneous conversations were recorded in 2012–2014 and contain speaker and context metadata.

BNC2014 serves as a transport target rather than a direct replication of `languageR::dative`. Its outcome field is `Pattern`, where the verb–noun–noun code `VNN` maps to NP realization and the verb–noun–prepositional-phrase code `VNPP` maps to PP realization. The six shared verbs are *give*, *send*, *show*, *sell*, *offer*, and *lend*. All six occur in the packaged `languageR` dataset.

4 Modern modelling update

The `languageR` result is a robustness result. The central question is whether modern modelling changes Bresnan et al.'s linguistic conclusion or mainly repackages it statistically. The answer, so far, is the latter. The table reports log loss, area under the receiver operating characteristic curve (AUC), and effective degrees of freedom (df).

Model	Test log loss	Test AUC	df
Marginal null	0.554	0.500	1
Non-verb main model	0.271	0.932	18
Fixed-verb full model	0.234	0.951	92
Hierarchical verb-intercept model	0.233	0.952	19

Table 1: Held-out `languageR::dative` results for the main model sequence.

The non-verb model already predicts held-out production choices well. Adding fixed verb effects improves test log loss from 0.271 to 0.234 and raises AUC from 0.932 to 0.951. The improvement confirms that lexical conditioning is not a minor residue after length, animacy, definiteness, pronominality, and accessibility are included.

The hierarchical verb-intercept model reaches almost the same held-out performance with far fewer effective parameters. Its test log loss is 0.233 and its AUC is 0.952. The estimated verb random-intercept standard deviation is 2.106, so the lexical signal remains large under partial pooling.

The null checks behave as they should. Scrambled-label fits lose predictive value. Fake-data fits without a verb signal collapse toward singular zero-variance random-effects models, while a fake verbeffect check recovers a nonzero verb standard deviation. These checks make the production result more credible, but they do not change its explanatory level.

The result supports Bresnan et al.'s production-choice conclusion. It does not create a new grammatical conclusion on its own. The same-data analysis shows that the classic profile is reproducible and statistically stable. The paper's new empirical pressure has to come from transport.

5 Generalization and acceptability

The BNC2014 analysis asks whether the production profile travels across corpus design, variety, period, and annotation regime. The transport model is trained on the spoken `languageR` rows restricted to the six BNC2014 verbs. It's then evaluated on harmonized BNC2014 rows, with complete-case filtering stated as part of the evidential scope.

Three estimands are kept apart. The `languageR` models estimate conditional production choice among attested noun phrase and prepositional phrase dative tokens. The BNC2014 transport result estimates out-of-domain predictive performance over complete-case rows in the six shared high-frequency verbs after harmonization. DAIS estimates controlled double-object preference for constructed alternatives. These numbers can therefore be read together only as different evidence channels, not as repeated measurements of one quantity.

Cross-variety probabilistic grammar of the dative isn't new ground. Bresnan and Ford (2010) compared American and Australian English and found that speakers' probabilistic choice knowledge is largely shared across varieties, but varies enough to surface in naturalness ratings and lexical-decision latencies. Röthlisberger et al. (2017) extended the comparison across nine post-colonial Englishes and found broad stability with local traces of indigenization. So the bare finding that dative conditioning largely travels wouldn't, on its own, be new.

Two things here are different. First, it's a strict out-of-domain test: one model is fit on one corpus and scored on another against held-out log loss, area under the curve, and scrambled-label nulls, rather than fit separately within each variety. Second, the transport result feeds a distinction the variationist work doesn't draw, between production probability and grammatical possibility, developed in the next section. The contribution is that framing and the test, not the stability itself.

The comparison is therefore an out-of-domain transport test. The first transport model holds modality close by training only on spoken `languageR` rows, but the corpus shift is still large: American telephone conversation from 1990–1991 to spontaneous British conversation from 2012–2014, under a different annotation scheme and a six-verb target set. That target set is already a scope restriction. In `languageR`, those six verbs account for 2,201 of 3,263 rows, and for 1,564 of 2,360 spoken rows. They are the high-frequency core, not a random sample of the alternation.

Harmonization is lossy. The outcome maps cleanly: `VNN` becomes NP and `VNPP` becomes PP. Verb also maps cleanly after restricting the `languageR` data to the six shared verbs. Length needs more care. The released BNC2014 length fields are character counts: the data dictionary defines `RecLen` and `ThemeLen` as the number of characters in the recipient and theme (G. B. Jenset & McGillivray, 2019). Those aren't comparable to `languageR`'s token-like lengths, so the released fields can't enter the transport model directly. The script instead derives word counts from the recipient and theme strings. Two costs follow: the whitespace tokenization isn't guaranteed to match the convention behind `languageR`'s counts, and the few rows with empty phrase strings drop out, since a character count can't be turned back into a word count.

Several predictors are available only as proxies. BNC2014 has recipient and theme animacy, theme definiteness, and recipient/theme pronominality, but some fields have missing values and different annotation paths. Recipient definiteness is not released as a field, so it enters only as a labelled regex proxy in a sensitivity check. Discourse accessibility is unavailable and is dropped from transport.

The core transport model therefore scores the 1,621 BNC2014 rows complete for the harmonized predictors, not all 1,839 released dative rows. The missing rows are not neutral: noun phrase recipients make up 0.819 of the complete rows but only 0.670 of the 218 incomplete rows. The headline comparison is consequently a complete-case estimate over the shared high-frequency core.

A BNC2014 metadata audit makes the domain label concrete. In the released dative rows, Level 1 dialect is "uk" for 1,720 of 1,839 rows and Level 2 dialect is "english" for 1,689. Gender is female-coded for 1,138 rows and male-coded for 699, and the largest age band is 19–29 with 736 rows. Interaction type is also skewed: close-family, partner, and very-close-friend conversations contribute 1,171 rows, while friends and wider-family conversations contribute another 538. The complete-case subset has the same shape. A single BNC2014 transport score should therefore be read as a corpus-domain diagnostic, not as a clean estimate for British spoken English in general.

Model	Test data	Log loss	AUC
`languageR` marginal	BNC2014 complete	0.473	0.500
`languageR` core transport	BNC2014 complete	0.308	0.906
`languageR` core transport	Scrambled BNC2014 complete	0.885	0.515
BNC2014 marginal holdout	BNC2014 complete	0.481	0.500
BNC2014 native core holdout	BNC2014 complete	0.202	0.960
BNC2014 native scrambled train	BNC2014 complete	0.495	0.440

Table 2: First BNC2014 complete-case transport and native holdout comparisons.

The transported core model beats the marginal baseline by a wide margin. Its test log loss is 0.308, compared with 0.473 for the `languageR` marginal baseline, and its AUC is 0.906. A percentile bootstrap over the BNC2014 complete-case test rows (2,000 resamples) puts the log loss in [0.277, 0.337] and the AUC in [0.890, 0.922], so the result isn't an artefact of one lucky split. When the BNC2014 outcome is scrambled, the same predictions collapse to 0.885 log loss and 0.515 AUC.

Simple all-row imputation checks make the complete-case scope visible. With missing lengths set to their BNC2014 medians and missing binary predictors set to their modes, the transported model scores 0.362 log loss and 0.872 AUC on all 1,839 rows. Filling the missing binary predictors uniformly as "no" or "yes" gives 0.312/0.911 and 0.408/0.856, respectively. The direction of the transport result survives, but the exact headline strength is sensitive to the missingness treatment.

Raw accuracy is a weak lens here. The recipient is an NP in about four of five complete-case BNC2014 tokens, so a majority-class baseline already scores near 0.82, and the transported model reaches only about 0.86. The gain that matters shows up in discrimination and calibration, the AUC of 0.906 and the lower log loss, not in the accuracy headline.

That framing changes the interpretation of the result. Success is evidence that the conditioning profile is projectible across the bundled corpus shift within the shared high-frequency core, not just

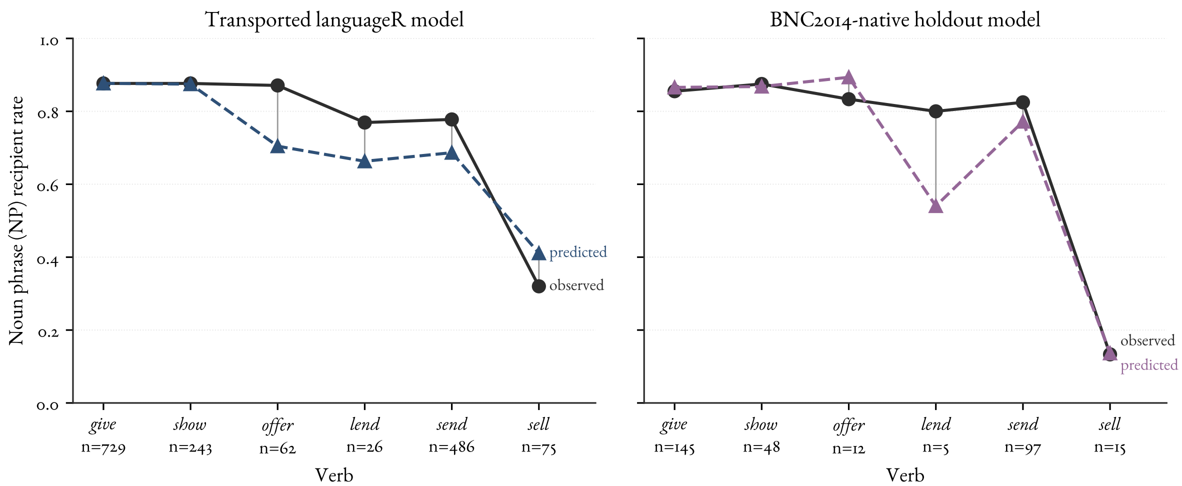

Figure 1: Verb-level calibration for the transported `languageR` model and the BNC2014-native holdout model. The vertical axis is the noun phrase (NP) recipient rate. Sample sizes are shown under each verb.

refittable. Failure, if it had occurred, would not have identified a single cause: variety, date, register, verb restriction, and annotation all move together.

A scope limit bounds how far this generalizes. The six shared verbs are among the most frequent and canonical members of the alternation, so transport is tested where the conditioning profile is most likely to hold. The two datasets intersect only on this high-frequency core; verbs with marginal, contested, or idiosyncratic dative behaviour aren't shared, so they go untested here. The result therefore licenses a claim about the core, not about the periphery, where projectibility is hardest to establish.

The BNC2014-native model is still stronger. On a verb-stratified BNC2014 holdout split, the native core model reaches 0.202 log loss (bootstrap [0.154, 0.254]) and 0.960 AUC (bootstrap [0.938, 0.979]). The gap is not small: the transported model's log loss is about half again the native model's, and the two bootstrap intervals don't overlap on either metric, so the difference isn't just split noise. The transported model has learned a useful profile, but BNC2014 also contains local structure that the `languageR` fit doesn't capture.

Read the gap as a bounded downgrade rather than an apology. The transported model clears the marginal and scrambled-label comparisons, so the result is not just a same-corpus artefact. But the residual gap cannot be assigned to variety, period, register, annotation, or the six-verb restriction one by one. Those dimensions are bundled in this comparison.

Several directions travel. Recipient length is negative for NP realization; theme length is positive; recipient pronominality is strongly positive; theme pronominality is strongly negative; `sell` and `send` are less NP-like than `give`. These are the strongest signs that the Bresnan profile is projectible beyond one packaged dataset.

Some details do not travel cleanly. The transported model underpredicts NP realization for `send`, `offer`, and `lend`, and overpredicts it for `sell`. Theme definiteness is the clearest coefficient mismatch: negative in the `languageR`-trained model, positive and weak in the BNC2014-native model.

The recipient-definiteness proxy should stay a sensitivity check. Adding it worsens transport log loss from 0.308 to 0.347 while leaving AUC essentially unchanged. That pattern fits the harmonization note: the proxy is interpretable, but it should not carry the headline result.

Acceptability evidence answers a different question. The Dative Alternation and Information Structure (DAIS) dataset, introduced by Hawkins et al. (2020b), contains 50,000 human judgements over 5,000 dative sentence pairs and systematically varies verb, definiteness, and argument length. The repository now carries a CC BY 4.0 licence, and the dataset authors have given explicit permission to reuse the judgements. DAIS can therefore enter as a compact bridge (Hawkins et al., 2020a).

The bridge keeps the targets separate. I score only the 150 DAIS items whose verbs overlap the six-verb production core. The same spoken `languageR` production model that was transported to BNC2014 predicts a noun-phrase recipient probability for each constructed DAIS pair; the DAIS item mean gives the corresponding human double-object preference on a 0–1 scale. As a bridge analysis, it asks whether a production probability profile and a controlled preference profile point in the same direction.

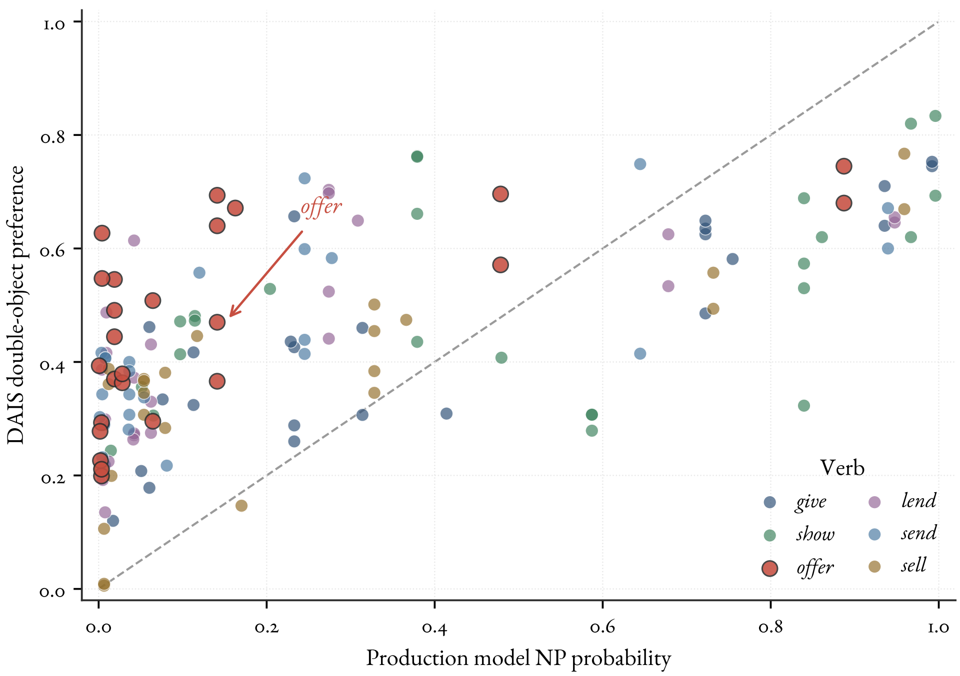

Figure 2: DAIS production/preference bridge for the six shared verbs. Each point is a constructed DAIS item. The horizontal axis is the spoken `languageR` model's noun phrase (NP) recipient probability; the vertical axis is DAIS double-object (DO) preference. The diagonal marks equality between the two channels, and the *offer* cluster is highlighted because it shows the clearest channel separation.

The two channels line up moderately, while remaining distinct. With *something* treated as a pronominal theme, Pearson's r is 0.667 and Spearman's ρ is 0.685 over the 150 shared-verb items. The mean production NP probability is 0.283, while the mean human DO preference is 0.440. Treating *something* as nonpronominal gives the same interpretation ($r = 0.645$, $\rho = 0.675$, mean production probability 0.343). The participant-level check points the same way without making the correlation

Verb	Prod. NP prob.	DAIS DO pref.	Difference
give	0.408	0.449	-0.041
lend	0.204	0.426	-0.222
offer	0.150	0.468	-0.319
sell	0.231	0.351	-0.120
send	0.196	0.428	-0.232
show	0.506	0.516	-0.009

Table 3: DAIS bridge for the six shared verbs. The DAIS column is the item-level mean double-object (DO) preference. The production column is the noun-phrase (NP) recipient probability from the spoken `languageR` core model.

carry the whole inference. A mixed model on 1,525 shared-verb judgements from 923 participants, with random intercepts for participant and verb, is non-singular; once recipient condition and theme type are included, the incremental production-score coefficient is small (0.052, standard error 0.060). The bridge is therefore evidence of shared conditioning and diagnostic divergence, not validation of one channel by the other.

The useful cases are the divergences. *Offer* is the clearest verb-level one: the production model makes it strongly PP-like, but DAIS participants are much more willing to choose the double-object version. For example, "Juan offered the woman a payment" has predicted NP probability 0.141 but DAIS DO preference 0.694. The point is channel separation: judgement preference can occupy a different region of the evidence space from corpus production probability.

6 Grammatical possibility

The empirical result supports a sharper vocabulary, not a shortcut from frequency to grammar. The `languageR` analysis shows that a classic production profile is reproducible under explicit validation. The BNC2014 analysis shows that much of that profile travels to the BNC2014 corpus-domain target within the shared high-frequency verb set. Both results concern production choice.

The target is grammaticality, a contested construct that I'll take as a community-conditioned licensing status for form–value, or form–meaning, relations (Nefdt, 2023; Pullum, 2019). Cognitive and social mechanisms stabilize that status across a speech community. The status is graded, not a clean threshold: a form belongs to the grammar to the degree that the relevant relation is licensed strongly enough to project to new cases, including cases that are rare, strongly conditioned, or absent from a finite corpus. That projection is the point. A category earns its keep by being projectible for a purpose (Boyd, 1999; Goodman, 1955), and the purpose here is predicting realization in unseen contexts and telling rare-but-grammatical cases apart from ungrammatical ones.

Acceptability preference, raw judgement, corpus frequency, formal generability, and standardness are neighbouring constructs. They can bear on licensing, but none fixes it alone. A corpus token records what a speaker or writer did in a context. A judgement task samples a noisy, role-dependent instrument: how a speaker, participant, or evaluator responds to constructed alternatives under controlled conditions.

The useful distinction is between target and channel. Corpus choices are one channel: they show which form was realized when an alternation token occurred. Judgements are another channel: they provide detector responses calibrated to a speech community and task role. Grammaticality is the target those channels bear on indirectly.

The DAIS bridge shows why the distinction matters. Its shared-verb judgements agree enough with the production model to show shared sensitivity to the dative conditioning profile. They also diverge enough to show that acceptability preference has a different evidential role from production probability. The *offer* cases are the useful warning: controlled judgements can make a double-object form look more acceptable than a corpus production model trained on attested choices would lead us to expect.

Everything here is conditioned on the opportunity set. This corpus evidence covers attested dative-alternation tokens: cases already annotated as NP or PP recipient realizations. It can support claims about which realization speakers chose inside that frame. A broader corpus design could estimate other opportunities: transfer events expressed by non-dative paraphrase, left implicit in context, or paired with an independent acceptability task. These released rows don't provide that broader denominator.

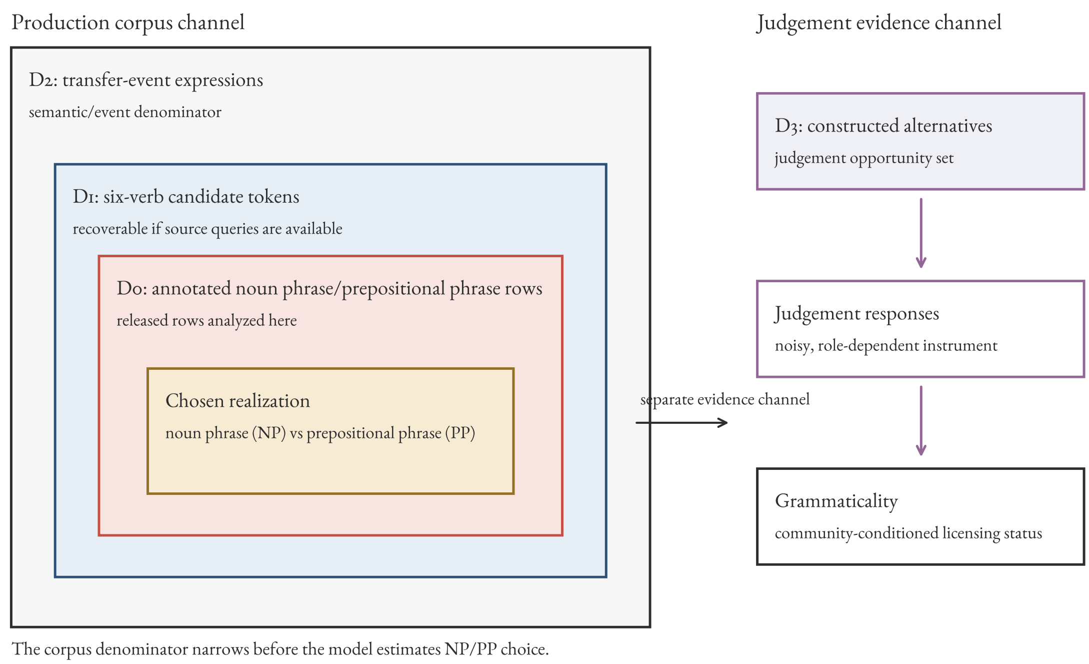

Figure 3: Nested opportunity sets for the dative-alternation evidence. The released corpus rows estimate realization choice inside the annotated noun phrase/prepositional phrase denominator. Broader transfer-event and judgement opportunity sets would answer different questions about grammaticality.

Production data can support claims about grammaticality only through an explicit bridge from attested choices to possible alternatives. Figure 3 marks where that bridge would have to be built.

The dative alternation makes the bridge visible. A low-probability double-object realization may be grammatical but dispreferred under the observed predictors. A prepositional realization may be likely because the theme is short and the recipient is long. A form may be absent because the corpus lacks the relevant verb, register, or discourse setting.

No amount of additional production data closes this gap. A production model estimates choice among realizations that occurred, and it's silent by construction about realizations that are grammatical but unattested, or grammatical outside the sampled frame. More tokens enlarge the attested region without ever reaching the unattested-but-grammatical one. The boundary between production probability and grammatical possibility isn't a defect of `languageR::dative` or BNC2014; it's a general constraint on what usage frequency can show about grammar.

In one sense this restates a familiar point: performance data underdetermine competence, and no finite corpus fixes the grammar (Chomsky, 1965; Miller & Chomsky, 1963). What the dative case adds is a measurable, channel-specific version of that boundary. The opportunity set says which cells a production channel can and can't reach, so the gap stops being a slogan and becomes a property of a stated dataset and design, open to the same kind of modelling the rest of the paper uses.

The current analyses therefore license a cautious claim. Bresnan et al.'s production-conditioning profile is durable and partly transportable within the shared high-frequency core. DAIS shows that the same profile is visible through judgements, while preference and production can separate. That supports the idea that corpus probabilities can reveal structure that informal judgements miss. It does not make production probability identical to grammaticality. What the durable, transportable profile does buy is prediction: it forecasts realization for unseen tokens within its scope of verbs, registers, and varieties, and it marks the low-probability cells where grammaticality has to be settled through another channel.

7 Conclusion

The reanalysis has two production results and one acceptability bridge. First, the `languageR` profile survives baselines, falsification checks, and partial pooling. The hierarchical model matches the fixed-verb model with a smaller effective parameterization, so modern modelling confirms the classic production-choice conclusion.

Second, the profile partly transports. A spoken six-verb model trained on `languageR` predicts BNC2014 complete-case rows well above a marginal baseline and fails against a scrambled BNC2014 outcome. The BNC2014-native model still performs better. The diagnostic verdict is bounded. Transport survives the null comparison in the high-frequency shared core, but the target corpus is internally structured and the comparison remains too bundled to identify separate variety, period, register, annotation, or missingness effects.

The bridge is DAIS. On the six verbs shared with the production core, human double-object preference tracks the production profile moderately, but the two channels diverge in diagnostically useful places. The participant-level check reinforces that reading: after recipient condition and theme type are included, the production score has only a small incremental association with individual judgements. The *offer* cases show why acceptability evidence cannot be replaced by a larger production corpus.

That combination is the paper's centre. The classic result is not overturned. It's made reproducible, tested against real nulls, and moved into a second corpus where it can succeed or fail. The success is strong enough to matter, and the failures are specific enough to guide the next analysis.

The conceptual lesson is the same one that motivates the title. Production probability is evidence about grammatical possibility, not a synonym for it. Open data, transport tests, and the DAIS bridge make that evidence stronger, but they also mark a structural boundary: a production model estimates choice within the attested frame and stays silent, by construction, about the grammatical-but-unattested region. Crossing that boundary needs acceptability evidence, not more tokens.

Data and code availability

The analysis scripts fetch public datasets to temporary locations. Derived files live in `data/derived/`; source and licence notes live in `notes/source-verification.md`. The DAIS item-level results are fetched to a temporary file from the CC BY 4.0 upstream repository, and the cleaned individual-judgement ZIP is fetched for the participant-level check. Both files are validated by MD5 and summarized only as derived bridge tables (Hawkins et al., 2020a). No raw DAIS files are committed.

Acknowledgements

The final artificial intelligence (AI) use disclosure will list the models used on the project.

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